

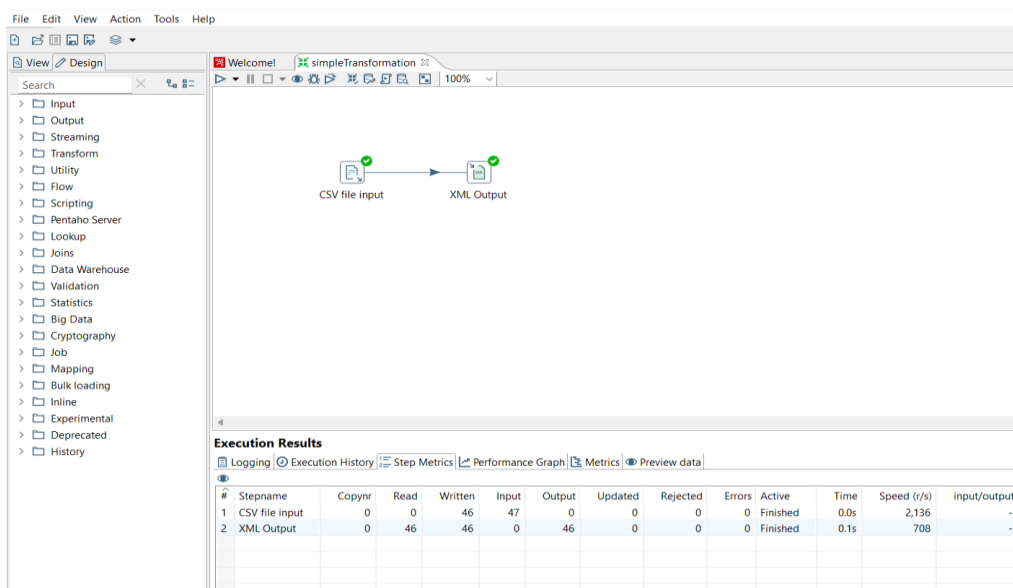
TP n°2 : Spoon - Transformation

Matière : Système d'information décisionnel

Ghada hamoudi
MPDS1

Atelier 1

Simple Transformation



The screenshot shows the Spoon IDE interface. On the left is a tree view of available components. The main workspace displays a 'Simple Transformation' job with two steps: 'CSV file input' and 'XML Output'. Below the workspace, the 'Execution Results' tab is active, showing a table with the following data:

#	Stepname	Copynr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	CSV file input	0	0	46	47	0	0	0	0	Finished	0.0s	2,136	-
2	XML Output	0	46	46	0	46	0	0	0	Finished	0.1s	708	-

```
person_data.xml X
C: > Users > ghada > OneDrive > Bureau > Master Data Science > person_data.xml
1  <?xml version='1.0' encoding='UTF-8'?>
2  <Rows>
3  <Row><Name>Ghada</Name> <City> Tunisia</City> <Job> data Scientist</Job></Row>
4  <Row><Name>John</Name> <City> Montreal</City> <Job> Salesman</Job></Row>
5  <Row><Name>Kamel</Name> <City> Lyon</City> <Job> Researcher</Job></Row>
6  <Row><Name>Nathalie</Name> <City> Montreal</City> <Job> Translator</Job></Row>
7  <Row><Name>Bush</Name> <City> Washington</City> <Job> President</Job></Row>
8  <Row><Name>Tim Bray</Name> <City> Vancouver</City> <Job> Technologist</Job></Row>
9  <Row><Name>Daniel</Name> <City> Montreal</City> <Job> Professor</Job></Row>
10 <Row><Name>Peter</Name> <City> Montreal</City> <Job> Professor</Job></Row>
11 <Row><Name>Jack</Name> <City> Montreal</City> <Job> Professor</Job></Row>
12 <Row><Name>Nathalie</Name> <City> Montreal</City> <Job> Professor</Job></Row>
13 <Row><Name>John</Name> <City> Montreal</City> <Job> Salesman</Job></Row>
14 <Row><Name>Kamel</Name> <City> Lyon</City> <Job> Researcher</Job></Row>
15 <Row><Name>Nathalie</Name> <City> Montreal</City> <Job> Translator</Job></Row>
16 <Row><Name>Bush</Name> <City> Washington</City> <Job> President</Job></Row>
17 <Row><Name>Tim Bray</Name> <City> Vancouver</City> <Job> Technologist</Job></Row>
18 <Row><Name>Daniel</Name> <City> Montreal</City> <Job> Professor</Job></Row>
19 <Row><Name>Peter</Name> <City> Montreal</City> <Job> Professor</Job></Row>
20 <Row><Name>Jack</Name> <City> London</City> <Job> Professor</Job></Row>
21 <Row><Name>Nathalie</Name> <City> Montreal</City> <Job> Professor</Job></Row>
22 <Row><Name>John</Name> <City> Montreal</City> <Job> Salesman</Job></Row>
23 <Row><Name>Kamel</Name> <City> Lyon</City> <Job> Researcher</Job></Row>
24 <Row><Name>Nathalie</Name> <City> Montreal</City> <Job> Translator</Job></Row>
25 <Row><Name>Bush</Name> <City> Washington</City> <Job> President</Job></Row>
26 <Row><Name>Tim Bray</Name> <City> Vancouver</City> <Job> Technologist</Job></Row>
27 <Row><Name>Daniel</Name> <City> Montreal</City> <Job> Professor</Job></Row>
28 <Row><Name>Peter</Name> <City> Montreal</City> <Job> Professor</Job></Row>
```

Atelier 2

Transformation avancée double

Spoon - simpleTransformation2

File Edit View Action Tools Help

View Design

Search

Input
Output
Streaming
Transform
Utility
Flow
Scripting
Pentaho Server
Lookup
Joins
Data Warehouse
Validation
Statistics
Big Data
Cryptography
Job
Mapping
Bulk loading
Inline
Experimental
Deprecated
History

simpleTransformation simpleTransformation2

CSV file input → Select values → XML Output

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

2024/03/13 21:46:21 - simpleTransformation2 - Dispatching started for transformation [simpleTransformation2]
2024/03/13 21:46:21 - XML Output.0 - Opening output stream in encoding: UTF-8
2024/03/13 21:46:21 - CSV file input.0 - Header row skipped in file 'C:\Users\ghada\OneDrive\Bureau\Master Data Science\person_data.txt'
2024/03/13 21:46:21 - CSV file input.0 - Finished processing (I=47, O=0, R=0, W=46, U=0, E=0)
2024/03/13 21:46:21 - Select values.0 - Finished processing (I=0, O=0, R=46, W=46, U=0, E=0)
2024/03/13 21:46:21 - XML Output.0 - Finished processing (I=0, O=46, R=46, W=46, U=0, E=0)
2024/03/13 21:46:21 - Spoon - The transformation has finished!!

```
<?xml version='1.0' encoding='UTF-8'?>
<Rows>
<Row><City> Tunisia</City></Row>
<Row><City> Montreal</City></Row>
<Row><City> Lyon</City></Row>
<Row><City> Montreal</City></Row>
<Row><City> Washington</City></Row>
<Row><City> Vancouver</City></Row>
<Row><City> Montreal</City></Row>
<Row><City> Montreal</City></Row>
<Row><City> Montreal</City></Row>
<Row><City> Montreal</City></Row>
<Row><City> Lyon</City></Row>
<Row><City> Montreal</City></Row>
<Row><City> Washington</City></Row>
<Row><City> Vancouver</City></Row>
<Row><City> Montreal</City></Row>
<Row><City> Montreal</City></Row>
<Row><City> Lyon</City></Row>
<Row><City> Montreal</City></Row>
<Row><City> Washington</City></Row>
<Row><City> Vancouver</City></Row>
<Row><City> Montreal</City></Row>
</Rows>
```

Spoon - simpleTransformation2

File Edit View Action Tools Help

View Design

Search

Input
Output
Streaming
Transform

- Add XML
- Add a checksum
- Add constants
- Add sequence
- Add value fields changing
- Calculator
- Closure generator
- Concat fields
- Get ID from slave server
- Number range
- Replace in string
- Row denormaliser
- Row flattener
- Row normaliser
- Select values
- Set field value
- Set field value to a constant
- Sort rows
- Split field to rows
- Split fields
- String operations
- Strings cut
- Unique rows
- Unique rows (HashSet)
- Value mapper
- XSL transformation

Utility
Flow
Scripting

simpleTransformation simpleTransformation2

100%

CSV file input

Unique rows

XML Output

Select values

Sort rows

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

2024/03/13 21:56:17 - Spoon - Launching transformation [simpleTransformation2]...

2024/03/13 21:56:17 - Spoon - Started the transformation execution.

2024/03/13 21:56:17 - simpleTransformation2 - Dispatching started for transformation [simpleTransformation2]

2024/03/13 21:56:17 - XML Output.0 - Opening output stream in encoding: UTF-8

2024/03/13 21:56:17 - CSV file input.0 - Header row skipped in file 'C:\Users\ghada\OneDrive\Bureau\Master Data Science\person_data.txt'

2024/03/13 21:56:17 - CSV file input.0 - Finished processing (I=47, O=0, R=0, W=46, U=0, E=0)

2024/03/13 21:56:17 - Select values.0 - Finished processing (I=0, O=0, R=46, W=46, U=0, E=0)

2024/03/13 21:56:17 - Sort rows.0 - Finished processing (I=0, O=0, R=46, W=46, U=0, E=0)

2024/03/13 21:56:17 - Unique rows.0 - Finished processing (I=0, O=0, R=46, W=7, U=0, E=0)

2024/03/13 21:56:17 - XML Output.0 - Finished processing (I=0, O=7, R=7, W=7, U=0, E=0)

2024/03/13 21:56:17 - Spoon - The transformation has finished!!

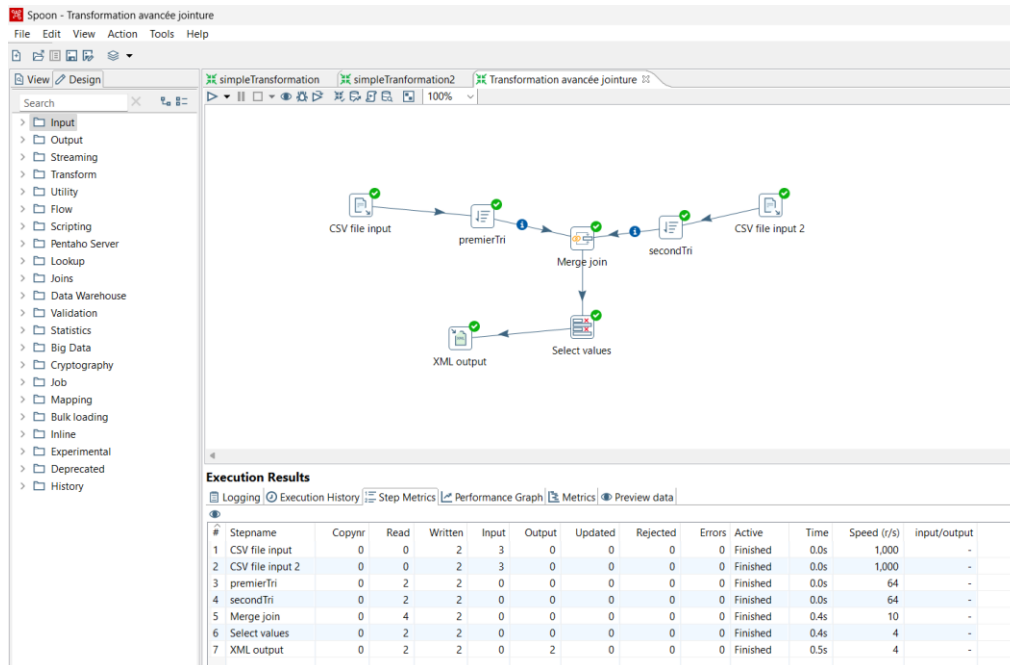
Fichier Edition Sélection Affichage ...

person_data2.xml X

C: > Users > ghada > OneDrive > Bureau > Master Data Science > person_data2.xml

```
1 <?xml version='1.0' encoding='UTF-8'?>
2 <Rows>
3 <Row><City> Dubai</City></Row>
4 <Row><City> London</City></Row>
5 <Row><City> Lyon</City></Row>
6 <Row><City> Montreal</City></Row>
7 <Row><City> Tunisia</City></Row>
8 <Row><City> Vancouver</City></Row>
9 <Row><City> Washington</City></Row>
10 </Rows>
11
```

Les jointures



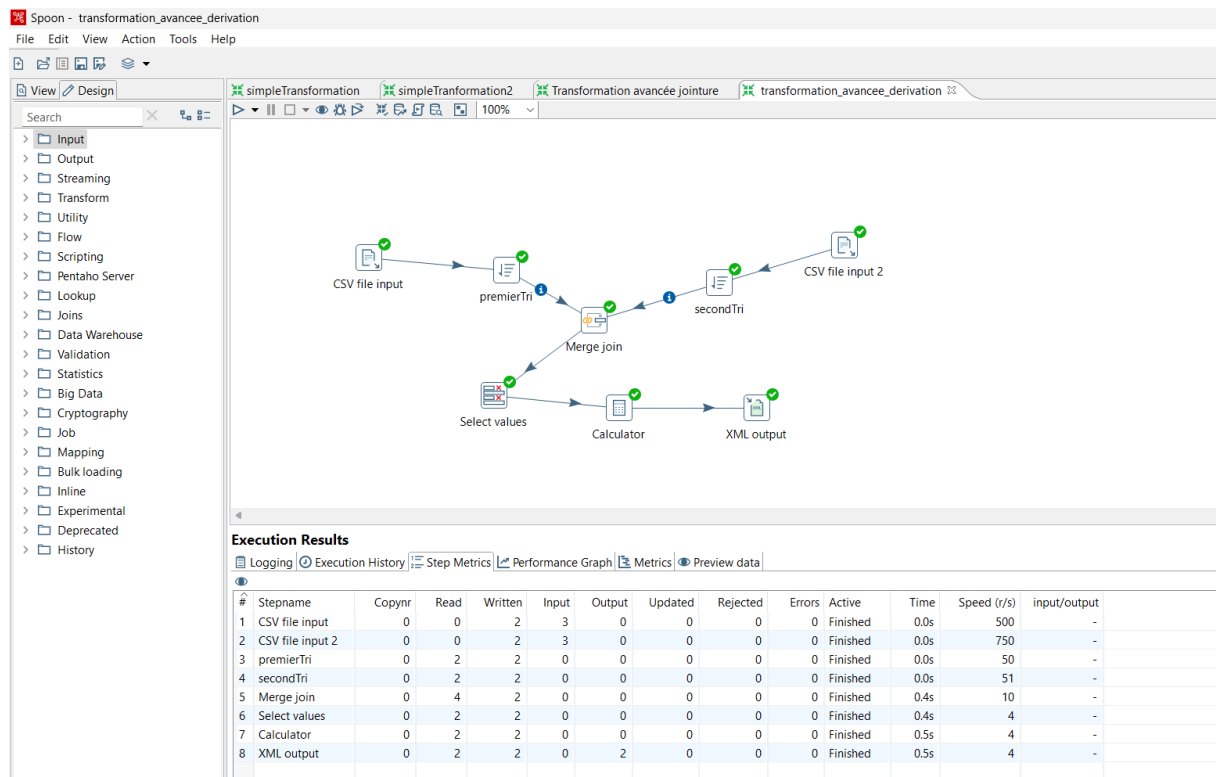
Fichier Edition Sélection Affichage ... Recherche

person_data4.xml.xml

C: > Users > ghada > OneDrive > Bureau > Master Data Science > person_data4.xml.xml

```
1 <?xml version='1.0' encoding='UTF-8'?>
2 <Rows>
3 <Row><nom>Pierre</nom> <ville>Drummondville</ville></Row>
4 <Row><nom>Jean</nom> <ville> Montreal</ville></Row>
5 </Rows>
6
```

Dérivation de nouveaux champs



```
person_data5.xml.xml X
C: > Users > ghada > OneDrive > Bureau > Master Data Science > person_data5.xml.xml
1 <?xml version='1.0' encoding='UTF-8'?>
2 <Rows>
3 <Row><ville>Drummondville</ville> <nom>Pierre</nom> <VILLE_1>DRUMMONDVILLE</VILLE_1></Row>
4 <Row><ville> Montreal</ville> <nom>Jean</nom> <VILLE_1> MONTREAL</VILLE_1></Row>
5 </Rows>
6
```

Opération Group By

The screenshot shows the Alteryx Designer interface. On the left is a tool palette with categories like Input, Output, Streaming, Transform, Utility, Flow, Scripting, Pentaho Server, Lookup, Joins, Data Warehouse, Validation, and Statistics. The 'Statistics' category is expanded, showing tools like Analytic query, Group by, Memory group by, Output steps metrics, Reservoir sampling, Sample rows, and Univariate statistics. The main workspace displays a workflow with four steps: 'CSV file input' (green checkmark), 'Sort rows' (green checkmark), 'Memory group by' (green checkmark), and 'XML output' (green checkmark). Arrows indicate the flow from input to sort, then to group by, and finally to output.

Execution Results

Logging | Execution History | Step Metrics | Performance Graph | Metrics | Preview data

#	Stepname	Copynr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	CSV file input	0	0	3	4	0	0	0	0	Finished	0.0s	1,333	-
2	Sort rows	0	3	3	0	0	0	0	0	Finished	0.0s	300	-
3	Memory group by	0	3	2	0	0	0	0	0	Finished	0.0s	187	-
4	XML output	0	2	2	0	2	0	0	0	Finished	0.0s	105	-

The screenshot shows a code editor with a dark theme. The top bar includes a search icon and the text 'Recherche'. Below the top bar, there are two tabs: 'person_data5.xml.xml' and 'person_data6.xml.xml.xml'. The active tab is 'person_data6.xml.xml.xml'. The editor shows the following XML content:

```
1 <?xml version='1.0' encoding='UTF-8'?>
2 <Rows>
3 <Row><City>Montreal</City> <TotalSales>20</TotalSales></Row>
4 <Row><City>Quebec</City> <TotalSales>20</TotalSales></Row>
5 </Rows>
6
```